

LAB 1: Measuring the Telescope Beamwidth with the Sun

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1. INTRODUCTION

Radio telescopes capture varying amounts of radiation in different directions. The temperature measured depends on the intensity of the radiation and on the sensitivity of the telescope, both of which are functions of sky position. The beam pattern of the telescope, or its sensitivity relative to position on the sky, can impact observations. Because of this, it is important to compare the radio telescope’s actual beam pattern with the theoretical beam pattern. To measure the angular width of the main beam, we calculate the half-power beam width (HPBW) or the full width between half-maximum points (FWHM).

```
: record lam_lab1_data_30
: freq 1430.0 1
: Sun
* Azimuth
: offset -24 0
:30
: offset -21 0
:30
: offset -18 0
:30
: offset -15 0
:30
: offset -12 0
:30
: offset -9 0
:30
: offset -6 0
:30
: offset -3 0
:30
: offset 0 0
:30
: offset 3 0
:30
: offset 6 0
:30
: offset 9 0
:30
: offset 12 0
:30
: offset 15 0
:30
: offset 18 0
:30
: offset 21 0
:30
: offset 24 0
:30
: roff
```

Figure 1. Observing Script

2. OBSERVATIONS

In order to collect our data, we used the Small Radio Telescope (SRT) on the Sterling Hall rooftop. The SRT used is a parabolic dish 2.3 meters in diameter with a beam width of 7.0° and a dual-axis azimuth/elevation mount. It uses a phasing-type single-sideband scanning receiver. We connected to the telescope via Microsoft Remote Desktop to run our observing script with the following detailed settings.

We measured the radio signal beginning at -24° offset from the Sun and ending at +24° offset, increasing by 3° each step for a total of 16 measurement points. We scanned along the Azimuth axis and chose a frequency of 1430.0 MHz using observing mode 1 (64 frequency channels). We originally integrated for 10 seconds, but increased to 30 seconds to generate at least 10 spectra. See Figure 1 in the Appendix for the complete observing script.

3. DATA ANALYSIS

Note: The Python code used for analysis is shown at the end in the Appendix (Section 5).

3.1. Data Preparation

We began by inspecting our output file in Visual Studio to better understand our data. Each offset position resulted in 11 spectra. We read in the file to ignore lines starting with ‘*’ as these lines contained information pertaining to the step in the observing script. Based on the fields dictated in the UW SRT Manual, we labeled each column with its corresponding variable (i.e., Field 0=time, Field 1 = azimuth(deg), Field 2 = elevation(deg), Field 3 = azimuth offset(deg), Field 4 = elevation offset(deg)). The first 5 rows of the dataframe are shown below in 2.

time	az	el	az_offset	el_offset	spectrum
str	f64	f64	f64	f64	list[f64]
"2025:253:16:44:19"	129.1	48.5	-24.0	0.0	[38.0, 48.0, ... 48.0]
"2025:253:16:44:22"	129.1	48.5	-24.0	0.0	[39.0, 49.0, ... 48.0]
"2025:253:16:44:24"	129.1	48.5	-24.0	0.0	[38.0, 47.0, ... 48.0]
"2025:253:16:44:27"	129.1	48.5	-24.0	0.0	[39.0, 49.0, ... 48.0]
"2025:253:16:44:30"	129.1	48.5	-24.0	0.0	[38.0, 48.0, ... 48.0]

Figure 2. Head of Dataframe

3.2. Azimuth Scans

In Figure 2 above, we see the head of our dataframe. Focusing in on the 'spectrum' column, each observation contains a list of 64 numbers, corresponding to the 64 frequency channels. For each spectrum list, we calculated the integrated intensity by taking the mean of that list. Since the values are discrete, calculating the integral is the same as taking the sum and dividing by the bandwidth. Then, at each offset position, we calculated the mean and standard deviation of the integrated intensity. Each dot represents the average intensity at each Azimuthal offset position while the error bars are the calculated standard deviation of the average intensity.

Figure 3 shows the angular offset from the Sun in degrees (x -axis) versus the integrated intensity, which represents the average brightness temperature in Kelvin (y -axis).

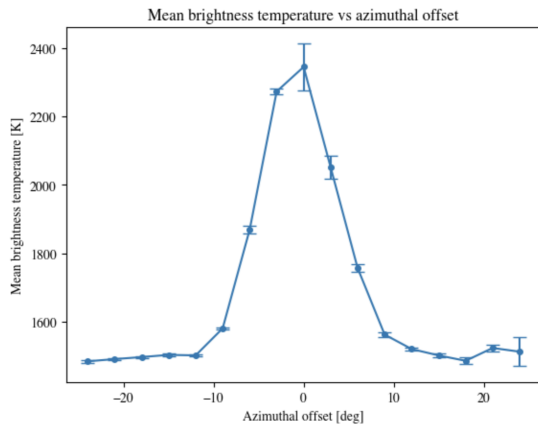


Figure 3. Azimuth Scan

3.3. Half-Power Beam Width (HPBW)

For the Azimuth scan, a 1 degree change in azimuth is only a 1 degree $\cosine$ (elevation) on the sky. To correct

the azimuth offset values, we multiplied the original offset by the cosine of the elevation. The plot below used similar methods as in Figure 3, but using the corrected azimuth offset. In Figure 4, we plotted a horizontal line at approximately 50% intensity (half power). Then, we plotted vertical lines and adjusted until they intersected with the beam pattern curve. The offset positions of these vertical lines are at $x = -3.7$ degrees and $x = 2.9$ degrees, yielding a distance between them of 6.6 degrees. Thus, our estimate of the full width at half-maximum (FWHM) or the half-power beam width (HPBW) is 6.6 degrees. This compares well the expected beamwidth for the SRTs which is around 7 degrees. The peak of the beam pattern does seem to be roughly centered around 0, with a slight fluctuation.

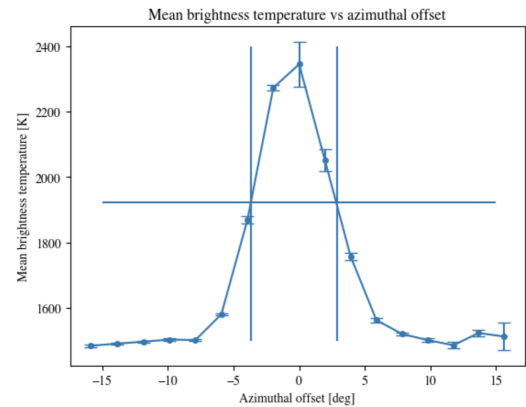


Figure 4. Half-Power Beam Width

3.4. Theoretical Beam Pattern

For a uniformly illuminated circular aperture with antenna dish of specified width, the expected beam pattern is given by

$$P(\theta) = \left(\frac{2J_1(x)}{x}\right)^2$$

where $x = \pi \sin(\theta)/(\lambda/D)$, θ is the angular offset in radians, D is the diameter of the SRT antenna, and $J_1(x)$ is the spherical Bessel function of the first kind with order 1. We wrote a Python function to calculate this using the telescope diameter of 2.3 meters. We used a frequency of 1430.0 MHz which we converted to wavelength using $c = f\lambda$, resulting in a wavelength value of 0.209645. Finally, we used the $j1$ function from the SciPy package. We scaled $P(\theta)$ by a constant (800) representing the height of the beam pattern curve and added a constant (1500) representing the minimum intensity recorded. The scaling ensures that the theoretical curve has the same peak value as the observed curve. Figure 5 shows the observed beam pattern (blue curve) along with the theoretical beam pattern (orange curve).

103 The shapes are fairly similar, although the theoretical
 104 curve is slightly right-shifted, narrower, and shorter than
 105 the observed curve.

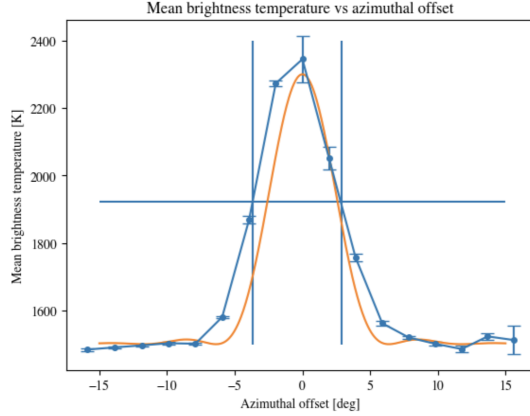


Figure 5. Theoretical vs Observed Beam Pattern

4. CONCLUSIONS

106 Overall, I think our data and calculations are close to
 107 what we would expect (within reason). Our calculated
 108 HPBW (6.6 degrees) is close to the true beam width
 109 of the SRT (approximately 7 degrees). The observed
 110 beam pattern was also close to the theoretical beam
 111 pattern, although our curve was a bit wider, taller, and
 112 left-shifted. One limitation is needing to record data at
 113 a proper time of day. Our first recording session was
 114 around 4:00 PM local time, so we encountered interfer-
 115 ence in our data. We decided to record new data the
 116 next day around 12:00 PM and saw almost no inter-
 117 ference. This worked out because we also learned to
 118 integrate over a longer time frame (30 seconds versus 10
 119 seconds) to get at least 10 spectra, since our first record-
 120 ing resulted in only 4 measurements per offset position.
 121 I noticed that the average intensity at 0° azimuthal offset
 122 has the largest error bar. I enjoyed this as a first lab to
 123 get accustomed to using radio telescopes.

5. APPENDIX

A. Python code.

Listing 1. Python Analysis Script

```

127 import numpy as np
128 import pandas as pd
129 import polars as pl
130 import matplotlib.pyplot as plt
131 from scipy.special import j1
132
133 plt.rcParams['font.family'] = 'Times'
134 plt.rcParams['mathtext.fontset'] = 'stix'
135
136 file = 'lam_lab1_data_30.rad'
137 with open(file, 'r') as infile, open('
138     radio_1_proc.rad', 'w') as outfile:
139     for line in infile:
140         if not line.startswith('*'):
141             line_proc = ','.join(line.
142                 split())
143             outfile.write(line_proc + '\n'
144                 )
145         else:
146             outfile.write(line)
147
148 data = pl.read_csv(
149     'radio_1_proc.rad',
150     comment_prefix='*',
151     has_header=False,
152     new_columns=['time', 'az', 'el', '
153         az_offset', 'el_offset', '
154         freq_1_bin', 'channel_width', '
155         spect_mode', 'num_channels']
156     + [str(x) for x in range(64)]
157 )
158
159 data = data.select(
160     pl.col('time'),
161     pl.col('az'),
162     pl.col('el'),
163     pl.col('az_offset'),
164     pl.col('el_offset'),
165     pl.concat_list(data.columns[9:]).alias
166         ('spectrum'),
167 )
168
169 data
170
171 agg_intensities = data.with_columns(
172     pl.col('spectrum').list.mean().alias('
173         integrated_intensity') #
174     Integrated intensity is
175     proportional to the mean of the
176     spectrum
177 )
178
179 ).group_by('az_offset').agg([
180     #
    Group measurements by offset

```

```

4:181 pl.col('integrated_intensity').mean().
182     alias('mean_intensity'), # Mean
183     intensity at each offset
4:184 pl.col('integrated_intensity').std().
185     alias('std_intensity'), # Stdev
186     intensity at each offset
4:187 ]).sort(
4:188     pl.col('az_offset')
4:189 )
4:190
4:191 agg_intensities
4:192
4:193 # figure 2
4:194
5:195 _x = np.linspace(-15, 15, num=100)
5:196
5:197 plt.errorbar(
5:198     agg_intensities['az_offset'],
5:199     agg_intensities['mean_intensity'],
5:200     fmt='-o',
5:201     capsize=5,
5:202     markersize=4,
5:203     yerr=agg_intensities['std_intensity']
6:204 )
6:205
6:206 plt.xlabel('Azimuthal offset [deg]')
6:207 plt.ylabel('Mean brightness temperature [K
208 ]')
6:209 plt.title('Mean brightness temperature vs
210 azimuthal offset')
6:211
6:212 plt.show()
6:213
6:214 # correct offset
6:215
7:216 corrected_offset = data.groupby('
217     az_offset').agg(
7:218     pl.col('el').mean()
7:219 ).with_columns(
7:220     (pl.col('az_offset') * pl.col('el').
221     radians().cos()).alias('corr_offset
222     ') # corr_offset <- offset * cos(El
223     )
7:224 ).sort('az_offset')
7:225
7:226 corrected_data = agg_intensities.join(
227     corrected_offset, on='az_offset')
7:228 corrected_data
7:229
7:230 # figure 3
8:231
8:232 plt.errorbar(
8:233     corrected_data['corr_offset'],
8:234     corrected_data['mean_intensity'],
8:235     fmt='-o',
8:236     capsize=5,
8:237     markersize=4,
8:238     yerr=corrected_data['std_intensity']
8:239 )

```

```

8:240 plt.hlines(0.5 * (corrected_data['
9:241     mean_intensity'].max() + 1500), -15,
242     15)
243
9:244 plt.vlines(-3.7, 1500, 2400)
9:245 plt.vlines(2.9, 1500, 2400)
9:246
9:247 plt.xlabel('Azimuthal offset [deg]')
9:248 plt.ylabel('Mean brightness temperature [K
249 ]')
9:250 plt.title('Mean brightness temperature vs
251     azimuthal offset')
9:252
9:253 plt.show()
9:254
10:255 # P(theta) function
10:256
10:257 def P(theta):
10:258     wavelength = 0.209645 # Observing
259     wavelength for a 1430.0 MHz
260     frequency, measured in meters
10:261     telescope_diameter = 2.3 # Measured in
262     meters
10:263
10:264     x = np.pi * np.sin(np.deg2rad(theta))
265     / (wavelength / telescope_diameter)
10:266     return (2 * j1(x) / x)**2
10:267
10:268 # figure 3
11:269
11:270 _x = np.linspace(-15, 15, num=500)
11:271
11:272 plt.errorbar(
11:273     corrected_data['corr_offset'],
11:274     corrected_data['mean_intensity'],
11:275     fmt='-o',
11:276     capsize=5,
11:277     markersize=4,
11:278     yerr=corrected_data['std_intensity']
12:279 )
12:280
12:281 plt.hlines(0.5 * (corrected_data['
282     mean_intensity'].max() + 1500), -15,
283     15)
12:284 plt.vlines(-3.7, 1500, 2400)
12:285 plt.vlines(2.9, 1500, 2400)
12:286
12:287 plt.plot(_x, 800 * P(_x) + 1500)
12:288
12:289 plt.xlabel('Azimuthal offset [deg]')
12:290 plt.ylabel('Mean brightness temperature [K
291 ]')
13:292 plt.title('Mean brightness temperature vs
293     azimuthal offset')
13:294
13:295 plt.show()
296

```